PHYSICS INFORMED ML MODELING FOR 1D CARBON TRANSPORT IN CARBURIZATION

Project report submitted to
Visvesvaraya National Institute of Technology, Nagpur
in partial fulfillment of the requirements for the award of the degree

Bachelor of Technology

In

Metallurgical and Materials Engineering

by

Sarthak Bondre (BT23MME018)
Yatharth Nema (BT23MME108)

under the guidance of

Prof. Dr. Abhinav Arya

Department of Metallurgical and Materials Engineering
Visvesvaraya National Institute of Technology
Nagpur 440 010 (India)

2025–26

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Declaration

We, Sarthak Bondre and Yatharth Nema, hereby declare that this project work titled

"PHYSICS INFORMED ML MODELING FOR 1D CARBON TRANSPORT IN CARBURIZATION"

is carried out by us in the Department of Metallurgical and Materials Engineering of Visvesvaraya National Institute of Technology, Nagpur. The work is original and has not been submitted earlier, whole or in part, for the award of any degree/diploma at this or any other Institution / University.

Date:21/04/2026

Sr.No.	Enrollment No	Names	Signature
1	BT23MME018	Sarthak Bondre	
2	BT23MME108	Yatharth Nema	

Certificate

This is to certify that the project titled **"PHYSICS INFORMED ML MODELING FOR 1D CARBON TRANSPORT IN CARBURIZATION"**, submitted by Sarthak Bondre and Yatharth Nema in partial fulfillment of the requirements for the degree of Bachelor of Technology in Metallurgical and Materials Engineering, VNIT Nagpur, is comprehensive, complete and fit for final evaluation.

Faculty Mentor
Metallurgical and Materials Engineering
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Head of Department,
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Abstract

Understanding carbon transport during carburization is essential for controlling microstructural evolution and mechanical properties in steels. Conventional diffusion models typically assume that carbon transport is governed solely by surface-driven diffusion; however, in realistic scenarios, internal processes such as carbide dissolution can act as additional sources of carbon within the material. These internal generation mechanisms are often difficult to quantify experimentally, leading to incomplete characterization of the carburization process.

In this work, a ***Physics-Informed Machine Learning (PIML)*** framework based on ***Physics-Informed Neural Networks (PINNs)*** is developed to model one-dimensional carbon transport in carburization, with particular emphasis on identifying internal carbon generation mechanisms. The governing diffusion equation is augmented with a source term representing localized carbon release, and the PINN is trained to satisfy the underlying physical laws, boundary conditions, and sparse observational data simultaneously. Unlike traditional numerical approaches, the proposed method does not rely solely on known inputs but is capable of solving an inverse modeling problem by inferring unknown source parameters within the system.

To validate the framework, synthetic reference data is generated using a finite difference solver for a controlled diffusion scenario with a prescribed internal source. A subset of this data is treated as sparse observations, mimicking realistic experimental measurements. The PINN successfully reconstructs the spatio-temporal carbon concentration field and accurately identifies the location and intensity of the internal source, demonstrating strong agreement with the reference solution. Error analysis indicates high predictive accuracy across the domain, with minor deviations observed near boundary regions and areas with steep concentration gradients. **Carburization provides the primary boundary-driven carbon influx, establishing the diffusion field, while the inverse modeling focuses on identifying additional internal carbon sources arising from carbide dissolution.**

The results highlight the potential of physics-informed machine learning as a powerful tool for solving inverse transport problems in materials science. This approach is particularly useful for analyzing carbon concentration profiles obtained from experimental techniques such as ***EPMA and SIMS***, enabling the inverse identification of hidden internal carbon sources in real carburization processes. This framework can be extended to real experimental datasets and more complex multi-physics systems, offering significant opportunities for process optimization and predictive modeling in advanced materials engineering. In particular, internal carbon generation due to carbide dissolution at elevated temperatures is modeled as a localized source.

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Nomenclature

Symbol	Description
x	Spatial coordinate (depth in material, normalized)
t	Time (normalized)
$C(x,t)$	Carbon concentration at position x and time t
$D(t)$	Diffusion coefficient (time-dependent)
$S(x,t)$	Internal carbon source term
x_0	Location of internal source (normalized depth)
A	Amplitude (strength) of source
σ_x	Spatial spread (standard deviation) of the source term
C_{FD}	Reference solution obtained using the finite difference method
C_{PINN}	Predicted concentration using Physics-Informed Neural Network
N_f	Number of collocation points used for PDE enforcement
N_i	Number of initial condition points
N_b	Number of boundary condition points
N_{data}	Number of observation (data) points
$\mathcal{L}$	Total loss function used during training
$\mathcal{L}_{\text{PDE}}$	Loss associated with PDE residual enforcement
$\mathcal{L}_{\text{data}}$	Loss due to mismatch with observed data
PINN	Physics-Informed Neural Network
FD	Finite Difference method

Introduction

1.1 Background

In recent years, artificial intelligence has emerged as a powerful tool for solving complex physical problems that were traditionally modeled using numerical or analytical methods. In geoscience and environmental systems, understanding transport processes such as contaminant dispersion, groundwater flow, and sediment transport is critical for effective resource management and pollution control. These processes are typically governed by partial differential equations (PDEs), which describe the evolution of physical quantities like concentration or temperature across time and space.

A closely related class of transport phenomena arises in materials engineering, particularly in heat-treatment processes such as carburization. Carburization is widely employed in the manufacturing of high-performance steel components, including gears, camshafts, and bearing races. It involves the diffusion of carbon from a carbon-rich atmosphere into the surface layers of a steel workpiece at elevated temperatures (typically 850°C–950°C), producing a hardened surface while maintaining a ductile core. The resulting carbon concentration gradient governs the final mechanical properties of the component, making accurate modeling of diffusion processes essential for process optimization and quality control.

Traditional numerical approaches such as the Finite Difference Method (FDM) and Finite Element Method (FEM) have been extensively used to solve these diffusion and transport equations, often based on Fick's law with temperature-dependent diffusivity described by Arrhenius relations. While these methods are reliable, they are computationally expensive, particularly for nonlinear, high-dimensional, or data-scarce systems. They also require fine meshing and stable discretization schemes, which can be challenging in irregular domains or under uncertain boundary conditions. Furthermore, classical approaches struggle with inverse problems, where unknown parameters or sources must be inferred from limited observations.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. (2019), offer an alternative framework by embedding the governing physical laws directly into the neural network's loss function. Instead of relying solely on large datasets, PINNs utilize PDEs along with initial and boundary conditions to guide the training process. Their mesh-free and differentiable nature makes them particularly suitable for problems involving complex geometries, incomplete or noisy data, and unknown parameters.

The present work applies a PINN framework to solve a one-dimensional, time-dependent

Advection–Dispersion Equation (ADE), a model frequently used in geoscientific and environmental studies to represent solute or contaminant transport through porous media. At the same time, the formulation draws parallels with diffusion-driven processes such as carburization, highlighting the broader applicability of PINNs across both environmental and materials engineering domains. The objective is to demonstrate that the PINN approach can approximate PDE-governed transport phenomena with high accuracy while maintaining computational efficiency and robustness in data-limited scenarios.

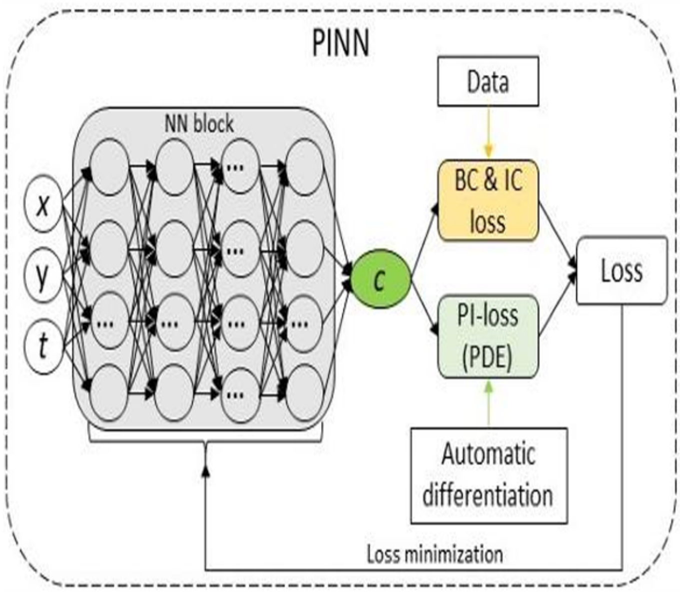

Figure 1.1: Architecture of the Physics-Informed Neural Network (PINN) used for carbon diffusion modeling.

1.2 Problem Statement

In many industrial carburization scenarios, the precise location and intensity of the carbon source within the furnace or at the material interface are not directly measurable. Instead, only spatially sparse, temporally distributed measurements of the carbon concentration field are available post-process. The challenge, therefore, is to reconstruct the unknown source term characterized by its spatial location and amplitude from these limited observations, while simultaneously ensuring that the recovered concentration field satisfies the underlying diffusion PDE.

Mathematically, the problem is formulated as follows. Let $C(x, t)$ denote the carbon concentration at depth $x \in [0, L]$ and time $t \in [0, T]$. The governing equation is:

$$\frac{\partial C}{\partial t} = D(t)\,\frac{\partial^2 C}{\partial x^2} + S(x, t; x_0, A) \tag{1.1}$$

where $D(t)$ is the temperature-dependent diffusivity following an Arrhenius relationship, and $S(x, t; x_0, A)$ is the source term parameterized by unknown source location x_0 and amplitude A. The inverse problem seeks to recover x_0 and A jointly with the concentration field $C(x, t)$, given sparse observations of C at selected (x, t) locations.

1.3 Objectives

The primary objectives of this study are:

- To develop and implement a Physics-Informed Neural Network (PINN) framework for the forward simulation of carbon diffusion in a one-dimensional normalized domain with a time-oscillating furnace temperature profile.
- To extend the PINN to an inverse formulation capable of simultaneously learning the neural network weights and two unknown physical parameters: the source location x_0 and source amplitude A.
- To validate the PINN predictions against a high-resolution finite difference reference solution, quantifying the accuracy of the recovered concentration field and the identified source parameters.
- To provide physical interpretations of the recovered source distribution in the context of carbide dissolution during carburization.

1.4 Theoretical Background and Modeling Framework

The transport of carbon during carburization is governed by diffusion processes within the material, combined with internal carbon generation mechanisms arising from microstructural transformations. In the present work, a one-dimensional formulation is adopted, where the carbon concentration $C(x, t)$ evolves as a function of spatial coordinate x and time t.

Unlike conventional forward simulations, the objective is not only to predict the concentration field but also to understand the underlying physical mechanisms contributing to carbon redistribution.

1.4.1 Governing Equation for Carbon Diffusion

The evolution of carbon concentration is described by the transient diffusion equation with an internal source term:

$$\frac{\partial C(x,t)}{\partial t} = D(t)\frac{\partial^2 C(x,t)}{\partial x^2} + S(x,t), \tag{1.2}$$

where:

- $C(x,t)$ is the carbon concentration,
- $D(t)$ is the diffusion coefficient,
- $S(x,t)$ represents internal carbon generation.

The first term on the right-hand side represents diffusion driven by concentration gradients, while the second term accounts for localized carbon generation within the material.

1.4.2 Physical Origin of Internal Carbon Source

The source term $S(x,t)$ represents carbon release due to carbide dissolution at elevated temperatures. During carburization, pre-existing carbides (e.g., Fe_3C) become thermodynamically unstable and dissolve into the surrounding matrix.

This dissolution process releases carbon atoms locally, leading to an increase in carbon concentration independent of surface diffusion. As a result, carbon transport is governed by both surface-driven carburization and internal generation.

1.4.3 Temperature-Dependent Diffusion

The diffusion coefficient is strongly dependent on temperature and is modeled using an Arrhenius-type relation:

$$D(t) = D_0 \exp\left(-\frac{E_a}{RT(t)}\right), \tag{1.3}$$

where:

- D_0 is the pre-exponential factor,
- E_a is the activation energy,
- R is the universal gas constant,
- $T(t)$ is the temperature.

This relation captures the exponential increase in diffusion rate at elevated temperatures.

1.4.4 Initial and Boundary Conditions

The system is solved under the following conditions:

$$C(x,0) = 0, \tag{1.4}$$

$$C(0,t) = 1, \tag{1.5}$$

$$C(1,t) = 0, \tag{1.6}$$

where:

- $C(x,0) = 0$ represents the initial low carbon content,
- $C(0,t) = 1$ represents a carbon-rich surface due to carburization,
- $C(1,t) = 0$ represents the low-carbon interior.

These conditions establish a concentration gradient that drives diffusion into the material.

1.4.5 Mathematical Representation of Source Term

The internal source is modeled using a Gaussian function:

$$S(x,t) = A \exp\left(-\frac{(x-x_0)^2}{\sigma_x^2}\right) \exp(-t), \tag{1.7}$$

where:

- x_0 is the location of carbon generation,
- A is the magnitude of the source,
- σ_x controls spatial spread.

This formulation represents localized and time-decaying carbon release.

1.4.6 Inverse Problem Formulation

The problem can be viewed as an inverse problem, where the internal source characteristics (x_0, A) are not known a priori.

Given observations of $C(x,t)$, the objective is to determine the source term $S(x,t)$ that best explains the observed concentration field.

This inverse formulation is central to understanding hidden carbon generation mechanisms within the material.

Methodology

The present study employs a hybrid computational framework combining numerical simulation and machine learning to model carbon transport and identify internal carbon sources. A Physics-Informed Neural Network (PINN) is used to approximate the solution of the governing diffusion equation while simultaneously solving an inverse problem to infer unknown source characteristics.

2.1 Problem Definition and Domain Setup

A one-dimensional domain is considered to represent carbon diffusion along the depth of the material. The spatial coordinate x is normalized such that:

$$0 \leq x \leq 1 \tag{2.1}$$

The temporal domain is defined as:

$$0 \leq t \leq t_{\text{final}}, \tag{2.2}$$

where $t_{\text{final}} = 0.4$ is selected to capture the transient diffusion regime.

The objective is to compute the carbon concentration field $C(x, t)$ and to identify unknown internal source parameters (x_0, A). This formulation enables the problem to be treated as an inverse modeling task.

2.2 Finite Difference Solver for Reference Data

A finite difference (FD) method is employed to generate a reference solution C_{FD}, which serves as a benchmark for validating the PINN model.

Discretization Approach

The governing diffusion equation is discretized in both space and time using a standard finite difference scheme. The spatial domain is divided into discrete nodes, and time is advanced incrementally using explicit updates.

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} = D(t)\frac{C_{i+1}^n - 2C_i^n + C_{i-1}^n}{\Delta x^2} + S(x_i, t_n) \tag{2.3}$$

Role of FD Solver

The FD solver plays a crucial role in this study:

- It provides a physically consistent reference solution C_{FD}
- It generates synthetic data for training and validation
- It enables controlled testing of the inverse modeling capability

Unlike experimental data, the FD solution ensures that the ground truth is known, allowing quantitative evaluation of model performance.

2.3 Data Generation and Sampling Strategy

The complete FD solution provides concentration values across the entire space–time domain. However, in realistic scenarios, only limited measurements are available.

Observation Data

A subset of N_{data} points is randomly sampled from the FD dataset:

- These points simulate sparse experimental measurements (e.g., EPMA or SIMS)
- They provide partial information about the concentration field
- They guide the inverse learning process

Collocation Points

To enforce the governing physics, N_f collocation points are generated across the domain:

- These points are not associated with data
- They are used to evaluate the PDE residual
- They ensure that the learned solution satisfies physical laws

This combination of sparse data and physics constraints is a defining feature of PINNs.

2.4 PINN Architecture

The carbon concentration $C(x,t)$ is approximated using a fully connected neural network.

Network Structure

- Input layer: spatial and temporal coordinates (x,t)
- Hidden layers: 4 layers with 64 neurons each
- Activation function: $\tanh$
- Output layer: predicted concentration C_{PINN}

The choice of $\tanh$ activation ensures smoothness and differentiability, which is essential for computing spatial and temporal derivatives using automatic differentiation.

Trainable Parameters

The model simultaneously learns:

- Neural network parameters (θ)
- Source parameters (x_0, A)

This enables the network to approximate both the solution field and the unknown source characteristics.

2.5 Loss Function Formulation

The training objective is defined through a composite loss function:

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{data}} \tag{2.4}$$

Each term enforces a different physical or observational constraint.

PDE Residual Loss

The PDE residual ensures that the predicted solution satisfies the governing diffusion equation:

$$\mathcal{L}_{\text{PDE}} = \text{MSE}\left(\frac{\partial C}{\partial t} - D(t)\frac{\partial^2 C}{\partial x^2} - S(x,t)\right) \tag{2.5}$$

Other Loss Components

- $\mathcal{L}_{\text{IC}}$: enforces the initial condition
- $\mathcal{L}_{\text{BC}}$: enforces boundary conditions
- $\mathcal{L}_{\text{data}}$: minimizes mismatch with observed data

The combined loss ensures that the solution is both physically consistent and data-driven.

2.6 Optimization Strategy

The model is trained using the Adam optimizer:

- Learning rate: 10^{-3}
- Training epochs: $\sim 10{,}000$

A relatively large number of epochs is required due to the complexity of the problem, as the model must simultaneously satisfy differential equations and perform parameter estimation.

During training:

- Gradients are computed using automatic differentiation
- Parameters are updated iteratively to minimize $\mathcal{L}$

Training convergence is monitored using the evolution of the loss function.

2.7 Inverse Modeling Framework

The central objective of this methodology is to solve an inverse problem. Instead of prescribing the source term, the model identifies it from available data.

- x_0 represents the location of internal carbon generation
- A represents the magnitude of the source

Physical Interpretation

The identified source corresponds to carbon release due to carbide dissolution. By reconstructing $S(x, t)$ from sparse data, the model enables identification of hidden internal mechanisms that cannot be directly measured.

This capability highlights the strength of PINNs in combining physical laws with data-driven learning for solving complex inverse problems in materials science.

Results and Discussion

3.1 Overview of Simulation Results

The developed Physics-Informed Neural Network (PINN) framework successfully modeled the transient carbon diffusion process and performed inverse identification of the internal carbon source. The results are validated against the reference solution obtained using the finite difference (FD) method.

In the FD solver, the internal source term $S(x,t)$ was defined using a Gaussian distribution with the source location deliberately fixed at:

$$x_0 = 0.4 \tag{3.1}$$

3.2 Training Point Distribution

Discussion

The distribution of training points highlights:

- Uniform coverage of the domain using collocation points (N_f)
- Boundary and initial condition enforcement along domain edges
- Sparse observation data distributed randomly

The spatial distribution of training points plays a crucial role in determining the accuracy and generalization capability of the PINN model. The use of uniformly distributed collocation points ensures that the governing differential equation is enforced across the entire domain, while the inclusion of boundary and initial condition points guarantees that physical constraints are satisfied at domain limits. Additionally, the randomly sampled observation data introduces localized information that guides the inverse learning process. This combination of physics-based and data-driven sampling enables the model to learn a globally consistent solution, even in regions where explicit data is sparse.

This combination ensures that the model learns both physical constraints and data-driven corrections.

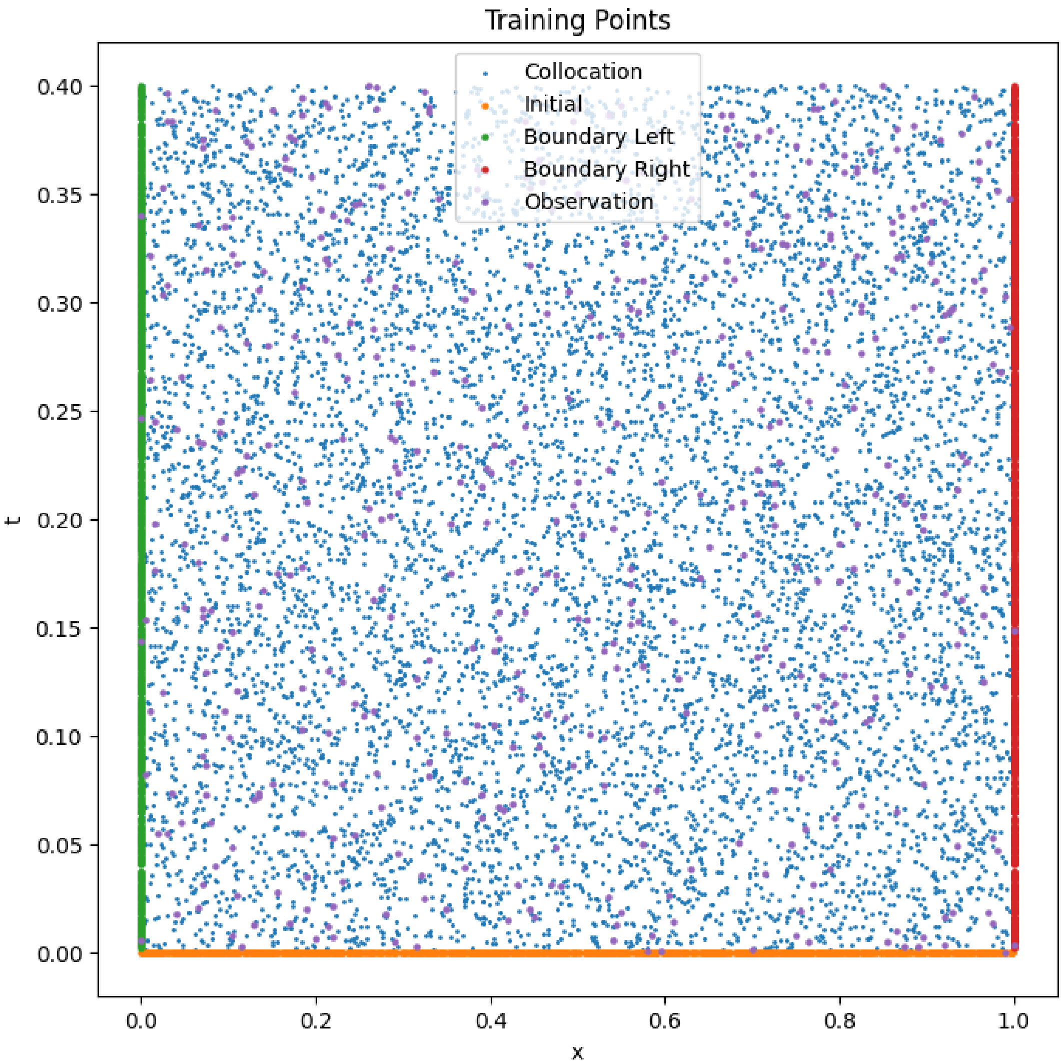

Figure 3.1: Distribution of collocation, boundary, initial, and observation points used for training.

3.3 Training Behavior and Convergence

Discussion

The loss curve demonstrates:

- Rapid initial decrease, indicating effective learning of dominant patterns.
- Gradual convergence as the model refines the solution.
- Oscillations due to competing loss terms.

The training behavior of the PINN model reflects the complex interplay between multiple loss components, including the governing equation, boundary conditions, and observational data.

During the initial stages of training, the model rapidly reduces the loss by capturing the dominant trends in the concentration field. As training progresses, finer adjustments are made to simultaneously satisfy all constraints, leading to a gradual convergence of the solution. The presence of oscillations in the loss curve indicates the competing nature of these objectives, particularly in regions with strong gradients or localized source effects. Nevertheless, the overall decreasing trend confirms that the model achieves stable convergence over extended training epochs.

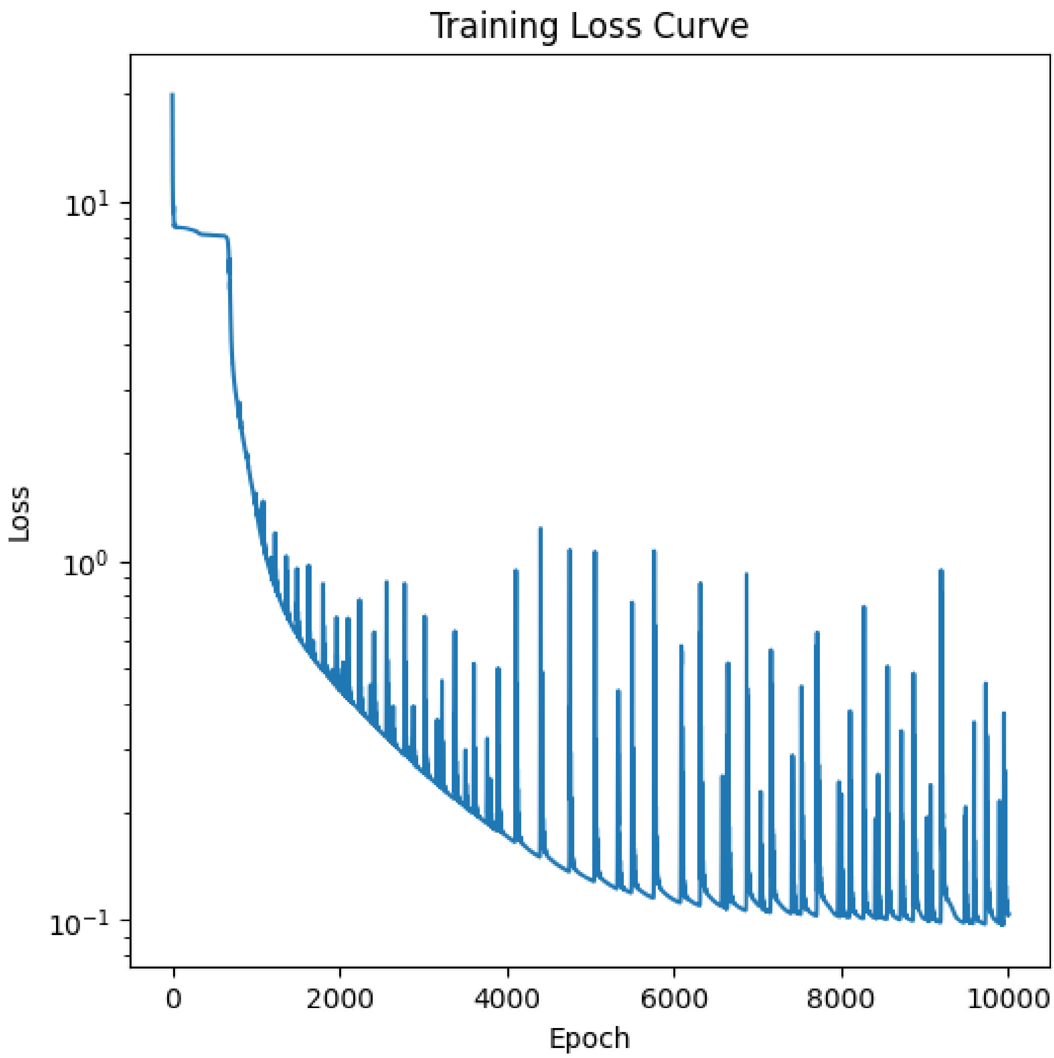

Figure 3.2: Training loss convergence over epochs.

3.4 Comparison of Carbon Concentration Profiles

Discussion

The comparison demonstrates strong agreement between PINN predictions (C_{PINN}) and FD reference solutions (C_{FD}).

- Peak concentration occurs near $x \approx 0.4$
- Temporal evolution is well captured
- Minor deviations near boundaries

The comparison of carbon concentration profiles at different time instances provides a direct assessment of the predictive capability of the PINN model. As diffusion progresses, the concentration profiles evolve from localized peaks to broader distributions, reflecting the spreading of carbon within the material. The presence of a consistent peak around $x \approx 0.4$ across all time steps indicates the influence of the internal source, which alters the classical diffusion behavior. The close agreement between the PINN and FD solutions demonstrates that the model successfully captures both the transient nature of diffusion and the localized impact of internal carbon generation.

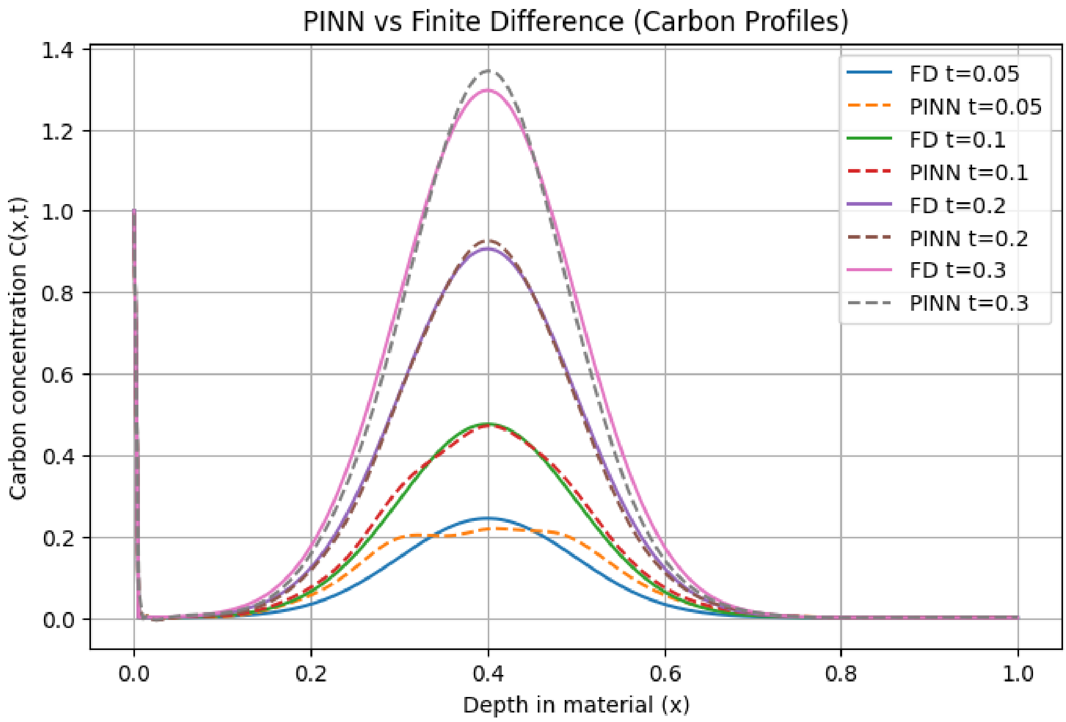

Figure 3.3: Comparison of carbon concentration profiles ($C(x, t)$).

3.5 Spatio-Temporal Distribution (Heatmap Analysis)

Discussion

The spatio-temporal evolution of the concentration field provides a comprehensive view of the diffusion process, capturing both spatial gradients and temporal progression simultaneously.

The formation of a high-concentration region around $x = 0.4$ highlights the dominant influence of the internal source, while the gradual spreading of contours with time reflects the diffusive transport governed by the time-dependent diffusion coefficient. The smooth and continuous nature of the PINN-generated contours indicates that the model has successfully learned a physically consistent representation of the solution across the entire domain.

- Clear concentration peak at $x = 0.4$
- PINN closely matches FD

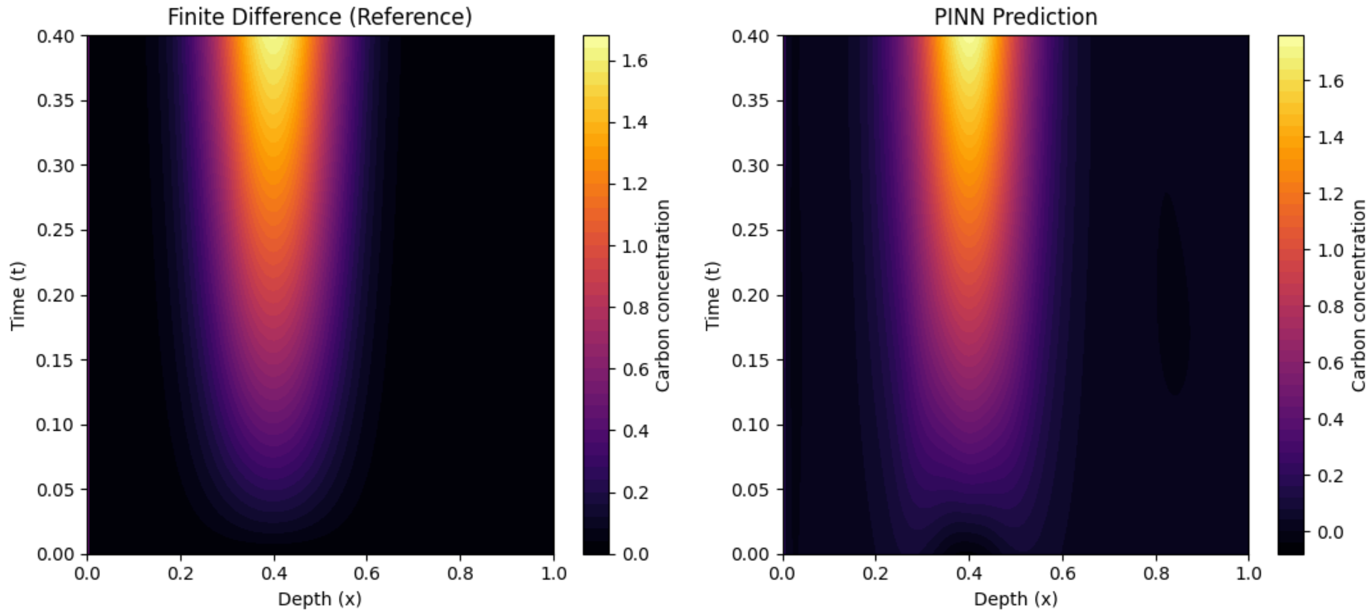

Figure 3.4: Spatio-temporal distribution (FD vs PINN).

3.6 Error Distribution Analysis

Discussion

The spatial distribution of error provides valuable insight into the regions where the model encounters the greatest difficulty in approximation. Higher errors near boundaries are primarily associated with steep concentration gradients imposed by boundary conditions, while localized discrepancies near the source region arise due to the sharp spatial variations introduced by the internal carbon generation. Despite these challenges, the error remains low across most of the domain, indicating that the PINN model maintains strong predictive accuracy and stability throughout the simulation.

- Low error across domain

- Higher error near boundaries

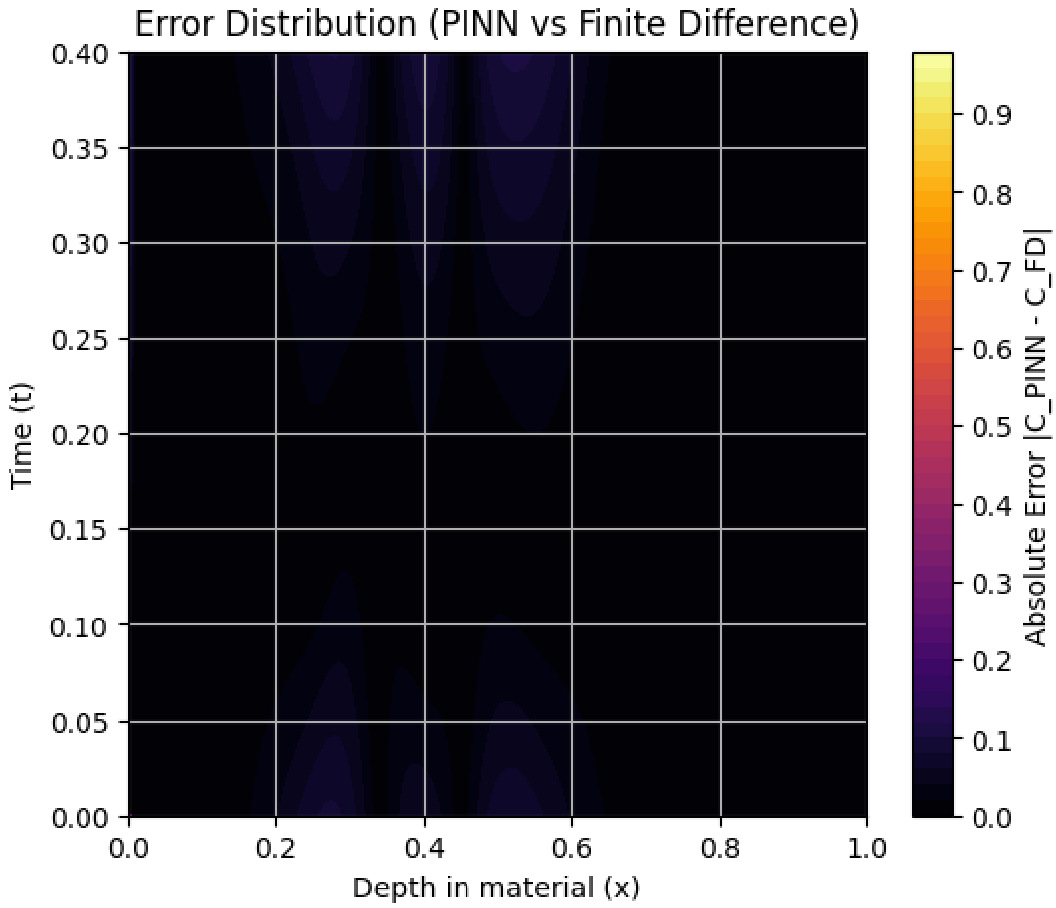

Figure 3.5: Error distribution between PINN and FD.

3.7 Recovered Source Term (Inverse Modeling)

Discussion

The recovered source shows a Gaussian profile centered at:

$$x_0 \approx 0.4 \tag{3.2}$$

- Peak $\rightarrow$ max carbon generation
- Width $\rightarrow$ spatial spread
- Amplitude $\rightarrow$ source strength

The reconstruction of the internal source term represents the most significant outcome of the inverse modeling framework. Unlike forward simulations where the source is predefined, the PINN model infers the source characteristics directly from observed data and governing physics. The resulting Gaussian profile centered near $x \approx 0.4$ confirms that the model has successfully identified the location of carbon generation. This capability highlights the strength of the approach in uncovering hidden internal mechanisms, such as carbide dissolution, which cannot be directly measured but significantly influence the overall diffusion behavior.

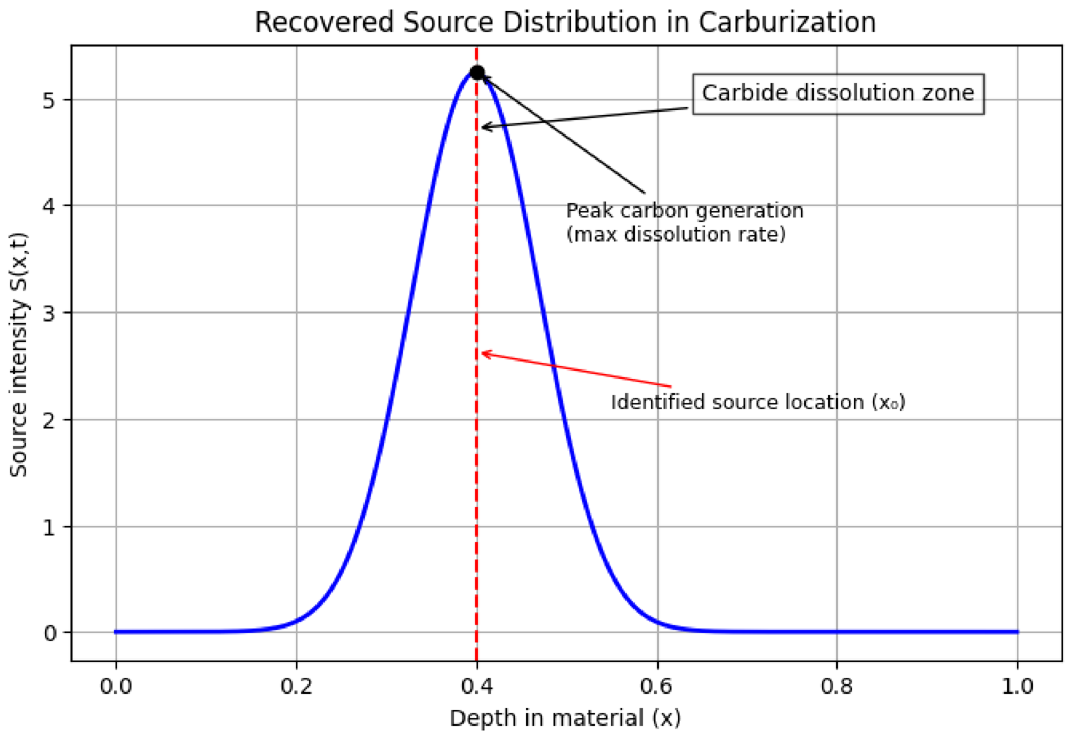

Figure 3.6: Recovered source distribution $S(x, t)$.

Conclusion

This project successfully developed a Physics-Informed Neural Network (PINN) framework for modeling one-dimensional carbon transport in a carburization environment and for solving an associated inverse problem. The approach combined classical diffusion physics with data-driven learning, enabling the prediction of the concentration field $C(x, t)$ while simultaneously identifying hidden internal carbon sources within the material.

The PINN model demonstrated strong agreement with the reference finite difference (FD) solution across multiple time instances. The spatial distribution and temporal evolution of carbon concentration were accurately captured, confirming that the governing diffusion equation and boundary conditions were effectively learned by the network.

A key highlight of this work is the successful reconstruction of the internal source term $S(x, t)$. In the FD solver, the source location was deliberately fixed at $x_0 = 0.4$, and the PINN framework was able to recover this location with high accuracy, along with the corresponding source strength. This validates the capability of the model to solve inverse problems and extract physically meaningful parameters from limited observations.

The reconstructed source term can be physically interpreted as carbon generation due to carbide dissolution at elevated temperatures. This establishes a direct connection between the mathematical formulation and underlying metallurgical phenomena, demonstrating that the model is not only predictive but also physically interpretable.

The major outcomes of the study can be summarized as follows:

- Accurate prediction of carbon concentration profiles using PINN, consistent with FD solutions.
- Successful inverse identification of internal source parameters (x_0, A).
- Validation of PINN as a reliable tool for solving coupled forward and inverse problems.
- Demonstration of how sparse data (N_{data}) combined with physics constraints (N_f) can yield accurate solutions.
- Physical interpretation of the source term as carbide dissolution within the material.

The training process, conducted over approximately 10,000 epochs, showed stable convergence behavior. The combination of collocation points, boundary enforcement, and observation data ensured that the solution remained both physically consistent and data-driven. Minor deviations were observed near boundary regions, which can be attributed to strong gradient effects and limited data coverage.

In summary, this work highlights the potential of Physics-Informed Neural Networks as a powerful framework for modeling transport phenomena and solving inverse problems in materials science. By integrating physics with machine learning, the approach enables the reconstruction of internal mechanisms that are otherwise difficult to measure experimentally.

Future Scope:

- Extension to two- and three-dimensional diffusion problems.
- Incorporation of realistic temperature profiles and thermodynamic models.
- Validation using experimental data such as EPMA or SIMS measurements.
- Exploration of more complex source formulations representing detailed carbide kinetics.

References

- M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” Journal of Computational Physics, 2019.
- A. Hennigh, “Lat-Net: Compressing lattice Boltzmann flow simulations using deep neural networks,” arXiv preprint, 2017.
- Y. Pang, W. Z. Meng, and G. E. Karniadakis, “PINNs for steady and unsteady heat conduction,” Computer Methods in Applied Mechanics and Engineering, 2019.
- G. E. Karniadakis et al., “Physics-informed machine learning,” Nature Reviews Physics, 2021.
- X. Meng et al., “PPINN: Parareal physics-informed neural network for time-dependent PDEs,” Computer Methods in Applied Mechanics and Engineering, 2020.
- K. Sun et al., “Neural network control of reaction–diffusion systems with long-term stability,” Nature Communications, 2022.
- D. K. Jaiswal, N. Kumar, and A. Kumar, “Analytical solution to advection–dispersion equation,” Journal of Water Resource and Protection, 2011.